AI ACADEMY · LEVEL 7 · GRADE 12 · WEEK 2 · STUDENT LESSON

Reading a research claim

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Reproducible AI Research Study

Before you start

Recall A question that evidence can answer. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

Your mission

> Start with the target: Explain the mechanism and terminology of reading a research claim. Try the recall before opening notes.

Your learning target

I can explain the **mechanism** and terminology of reading a research **claim**.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from A question that **evidence** can answer. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of reading a research **claim**

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

Retrieve what you already know

Recall A question that **evidence** can answer. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a **claim**.

> **My starting explanation**

This week's end goal

Explain reading a research **claim** using the records and boundaries of this fictional investigation.

STUDENT LESSON · 2 OF 18

Notice the evidence

> Investigate the supplied Reading a research claim case. Separate observations, given facts and assumptions before choosing an explanation.

Reading a **claim** requires its method, data, comparison and uncertainty. Titles and abstracts may omit conditions crucial to interpretation.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

> **List the given facts and one fact that is not supplied.**

STUDENT LESSON · 3 OF 18

See how it works

> Follow the mechanism for Reading a research claim. Connect each labelled input, deciding step and result.

Separate the **claim** from supplied **evidence**. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those **observation**s.

Reading a research claim — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases.	Separate the claim from supplied evidence. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those observations.	B has75% versus62.5%, a12.5-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

STUDENT LESSON · 4 OF 18

Compare the cases

> Compare the supplied cases for Reading a research claim. Identify the detail that changes the conclusion.

Compare a familiar case with a changed case

A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Keep the procedure fixed. Explain what changed in the inputs and result.

Change one thing in the pictured case

Change one thing

Change: The headline changes to "every learner improves twice as much".

Predicted result

Prediction: That learner **claim** is unsupported by these answer counts.

Why the result follows

Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

STUDENT LESSON · 5 OF 18

Words that unlock the idea

Use the vocabulary of Reading a research claim to explain a concrete decision, rather than

> only memorising definitions.

Word	Meaning
Claim	A statement offered as a conclusion
Evidence scope	The conditions represented by observations
Extrapolation	Extending beyond measured conditions

Use two terms to explain a step in this week's example.

STUDENT LESSON · 6 OF 18

Follow a worked example

> Follow the worked trace for Reading a research claim, predict a step, then compare with its explanation.

A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases.

B has75% versus62.5%, a12.5-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

Worked explanation · Reading a research claim

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

> The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

Learn from the worked case

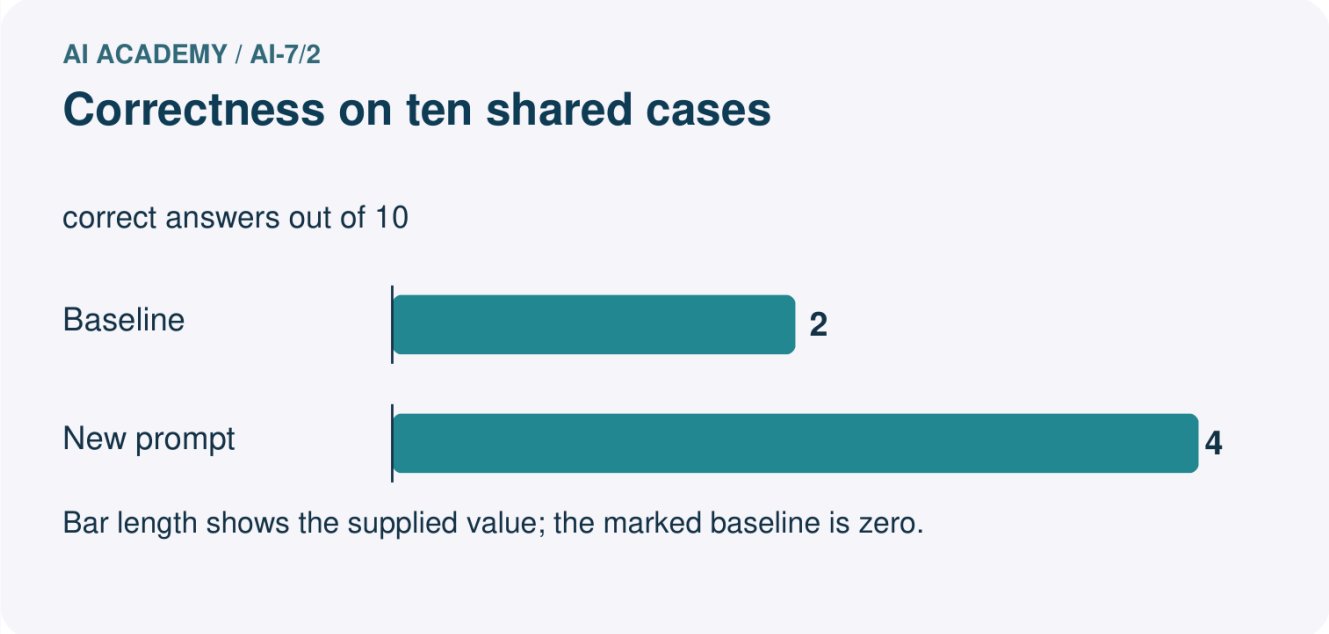
Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Reading a research claim

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

The facts we will use

> **Supplied facts**
>
> Baseline: 2/10 correct. New prompt: 4/10 on the same ten cases. Scoring is fixed.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-7/2

Reading a research claim

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Find the measured outcome

The table counts correct answers, not learning or satisfaction.



2. Calculate both changes

20% -> 40% is twice the rate and a 20-percentage-point increase.



3. Check claim scope

These ten cases support a descriptive comparison, not universal doubling.

CHANGE ONE THING

The headline changes to "every learner improves twice as much".

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Find the measured outcome	The table counts correct answers, not learning or satisfaction.
2. Calculate both changes	20%→40% is twice the rate and a 20-percentage-point increase.
3. Check claim scope	These ten cases support a descriptive comparison, not universal doubling.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

Why the result follows

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

What this example does not establish

Keep the stated limit

Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

Keep the stated limit

Read the labelled picture: Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.

Keep the stated limit

Change one thing: The headline changes to "every learner improves twice as much". Predict the result before checking the explanation. What remains the same?

STUDENT LESSON · 7 OF 18

Find and repair a mistake

> Find the unsupported step in the Reading a research claim claim and explain a repair using the supplied evidence.

A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Locate the earliest unsupported step

A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

State one repair and the checks you would rerun.

STUDENT LESSON · 8 OF 18

Practise together

> Practise Reading a research claim with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.

Explain your prediction to a partner before comparing results.

Practise with less help: Trace the worked case

Use workbook W2 for the connected example “Reading a research **claim**”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Check your reasoning

Use the worked case for Reading a research **claim**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the picture

Calculate the baseline and new rates for 2/10 and 4/10. Explain both twice the rate and the percentage-point change, and say why neither measures every learner’s improvement.

My explanation using a pictured step and the given facts

This is supported practice. Keep the separately named independent case for an attempt before feedback.

STUDENT LESSON · 9 OF 18

Run the investigation

> Investigate Reading a research claim. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

Predict, investigate and preserve the evidence

A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases. Show the intermediate steps and give the conclusion with its scope.

Record	Your evidence
Given input and fixed rule	
Prediction before checking	
Actual result, or “not run”	
First mismatch and one explanation	
Boundary or missing input and expected response	

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

Plan → try → check

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

STUDENT LESSON · 10 OF 18

Build your understanding

> Move from recall to tracing, diagnosis and transfer in Reading a research claim. Explain why each step is justified.

Explain → apply → challenge

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A numerical improvement can be real in the sample while the broader **claim** remains unsupported.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

Connect your project

> Apply this week's idea to Reproducible AI Research Study. Keep an annotated decision and the evidence supporting it.

Add a labelled reading a research **claim evidence** sheet to Reproducible AI Research Study. Show how you can explain the **mechanism** and terminology of reading a research **claim**.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Project evidence	Record
Component and version	
Inputs permitted at prediction time	
Trace and comparison with earlier work	
Failure, limit and human review	

Evidence for your project

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

STUDENT LESSON · 12 OF 18

Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

> **Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

> **My limited claim and a useful next check**

Study a complete case

> Revisit the complete Reading a research claim case for supported practice. Repeated examples do not become fresh assessments.

Complete evidence record

Reading a research claim — evidence trace

INPUT AND SCOPE	CHECK OR CALCULATION	CONCLUSION AND LIMIT
A fictional report says method B is better after6/8 supported answers versus baseline5/8 on the same constructed cases.	Separate the claim from supplied evidence. Record dataset, unit, baseline, metric and limitations. Ask whether the statement extends beyond those observations.	B has75% versus62.5%, a12.5-point difference on eight cases. Better in this measured outcome does not establish broader tasks or uncertainty.

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

My complete trace of the worked case

Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

Independent case: Apply a new case independently

Report **claim**s universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Attempt this without hints

Report **claim**s universal speed after median2 seconds versus3 on four local fixtures. Audit scope.

> **My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

STUDENT LESSON · 15 OF 18

Your lesson route

> Use this map to navigate Reading a research claim, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

By the end, I can	How I will show it
Explain the mechanism and terminology of reading a research claim	Explain the deciding step, show a trace or test, and state one limit.
Apply the stated procedure to a new case with a traceable result	Explain the deciding step, show a trace or test, and state one limit.
Identify an assumption or limitation and design a revealing follow-up	Explain the deciding step, show a trace or test, and state one limit.

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

STUDENT LESSON · 16 OF 18

Finish and return

> Explain Reading a research claim without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of reading a research **claim**

> **My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

STUDENT LESSON · 17 OF 18

Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? These ten cases support a descriptive comparison, not universal doubling.

1. Check **claim** scope

2. Calculate both changes

3. Find the measured outcome

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Check **claim** scope These ten cases support a descriptive comparison, not universal doubling. A research **claim** compresses a method and result.

Unpack the measured variable, denominator and comparison before accepting the headline.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: The headline changes to "every learner improves twice as much". Which conclusion is supported by the supplied facts?

1. We can decide the result without examining the changed **input** or the procedure.

2. This one example guarantees correct results for every future **input**.

3. That learner **claim** is unsupported by these answer counts.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That learner **claim** is unsupported by these answer counts. The outcome and experimental unit have changed in the wording without new **evidence**.

Repair a missing building block

Review A question that **evidence** can answer (ai-7/1). Goal: Explain the **mechanism** and terminology of a question that **evidence** can answer. Short practice: Which proposed diagram question has a measurable outcome?

Name its versions, six-case denominator and fixed scoring, then explain why giving only B easier tasks spoils the comparison.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

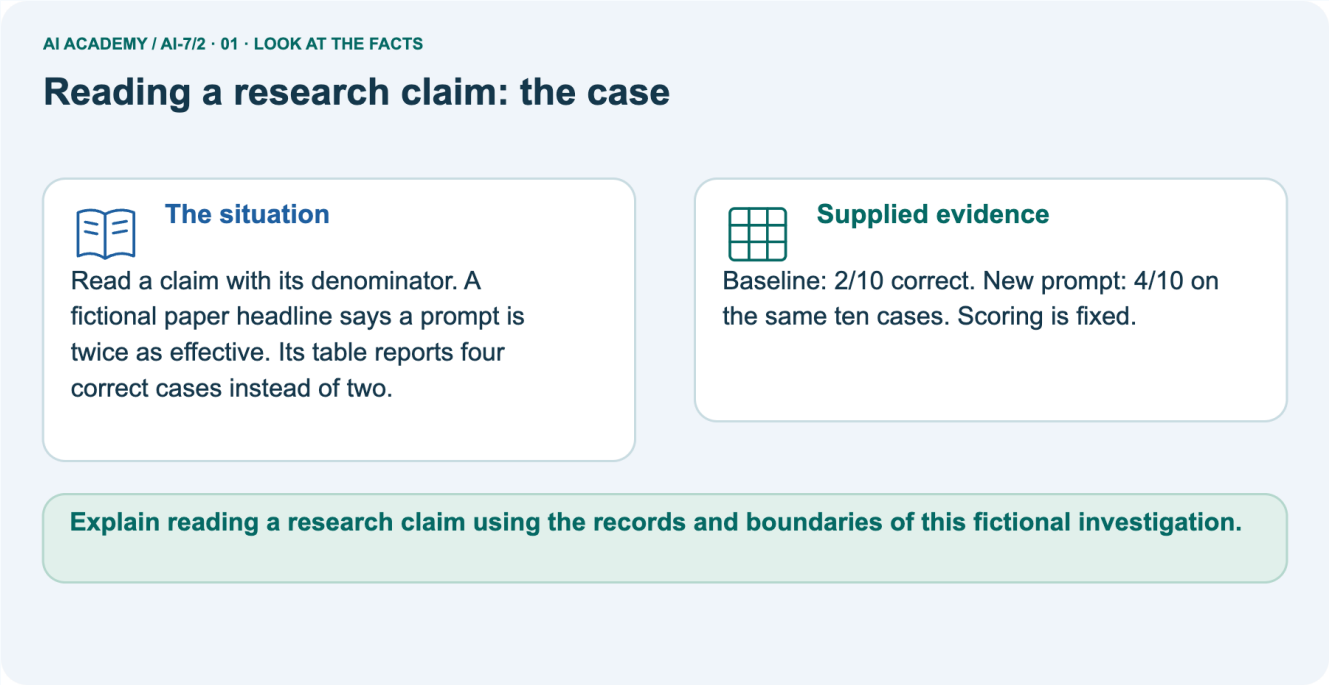
Drafts stay in this browser when storage is available; clear drafts on a shared device.

STUDENT LESSON · 18 OF 18

Visual concept journey

> Explain each Reading a research claim graphic using its labels, deciding evidence and relevant limitation.

01. Reading a research claim: the case



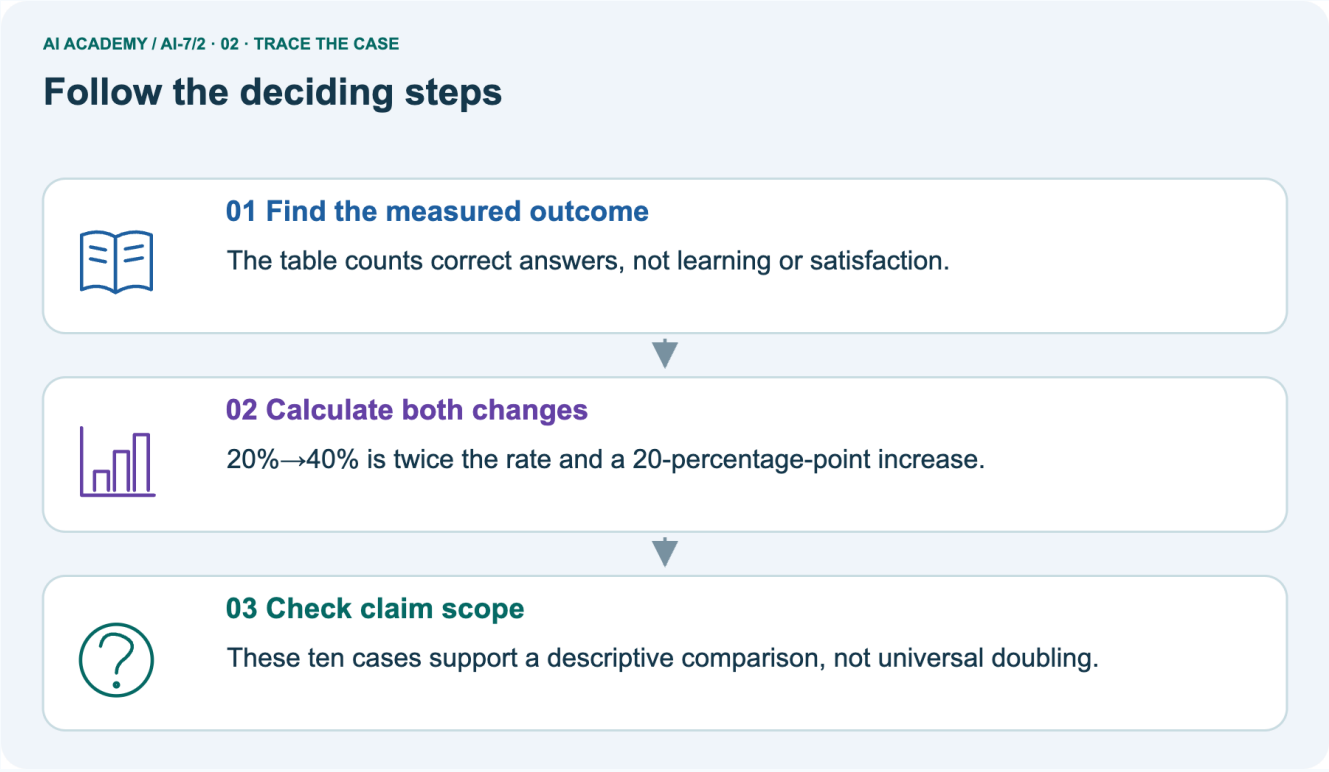
Follow the labelled picture with the matching explanation. Open the vector diagram

Read a **claim** with its denominator. A fictional paper headline says a prompt is twice as effective. Its table reports four correct cases instead of two. Supplied facts: Baseline: 2/10 correct.

New prompt: 4/10 on the same ten cases. Scoring is fixed.

> Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

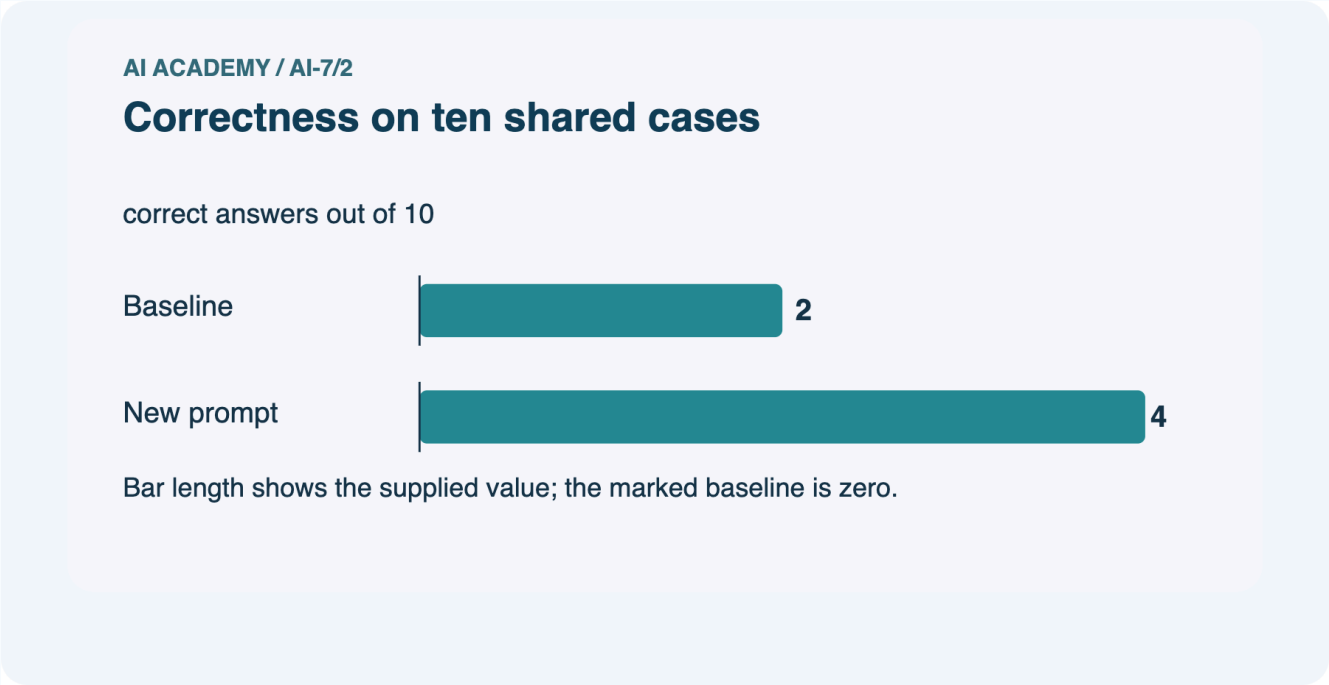


Follow the labelled picture with the matching explanation. Open the vector diagram

Find the measured outcome: The table counts correct answers, not learning or satisfaction. → Calculate both changes: 20%→40% is twice the rate and a 20-percentage-point increase. → Check **claim** scope: These ten cases support a descriptive comparison, not universal doubling.

> Point to each numbered stage and explain which recorded **input** it uses.

03. Correctness on ten shared cases

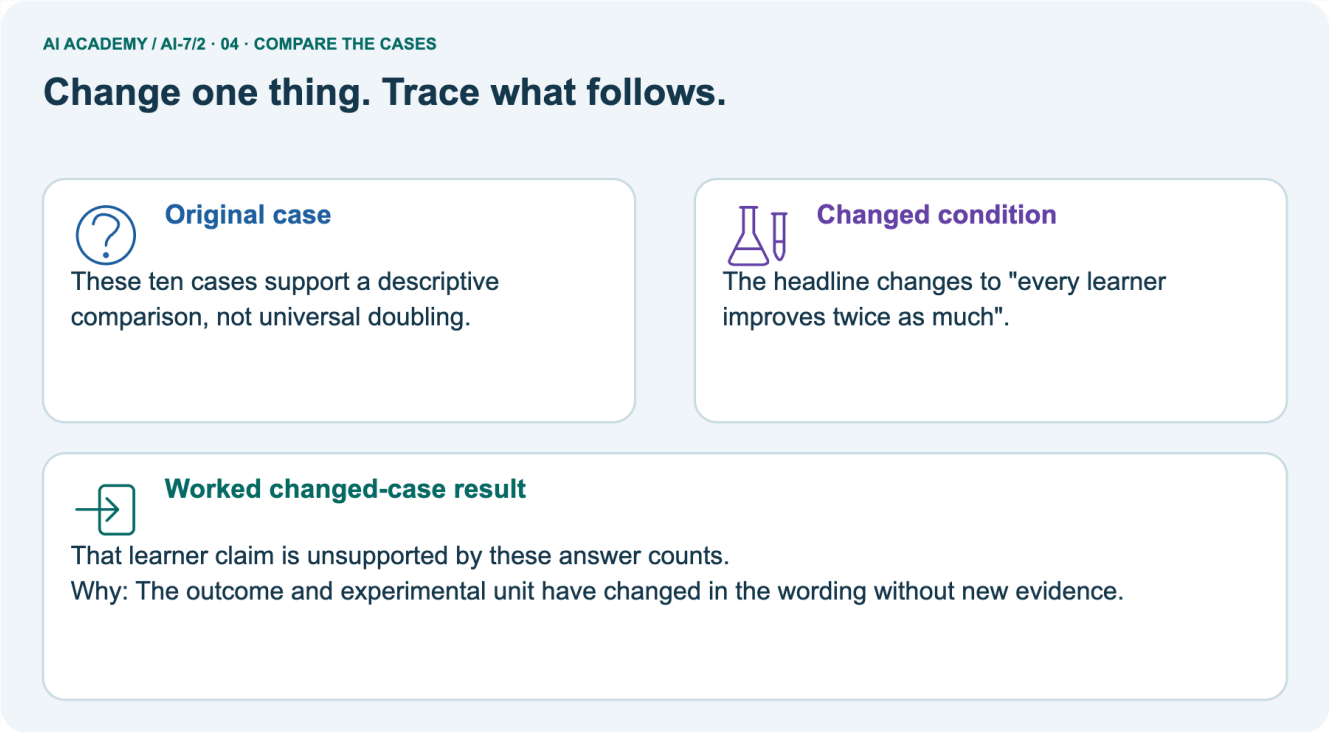


Follow the labelled picture with the matching explanation. Open the vector diagram

Correctness on ten shared cases. Units: correct answers out of 10. Baseline: 2. New prompt: 4.

> Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



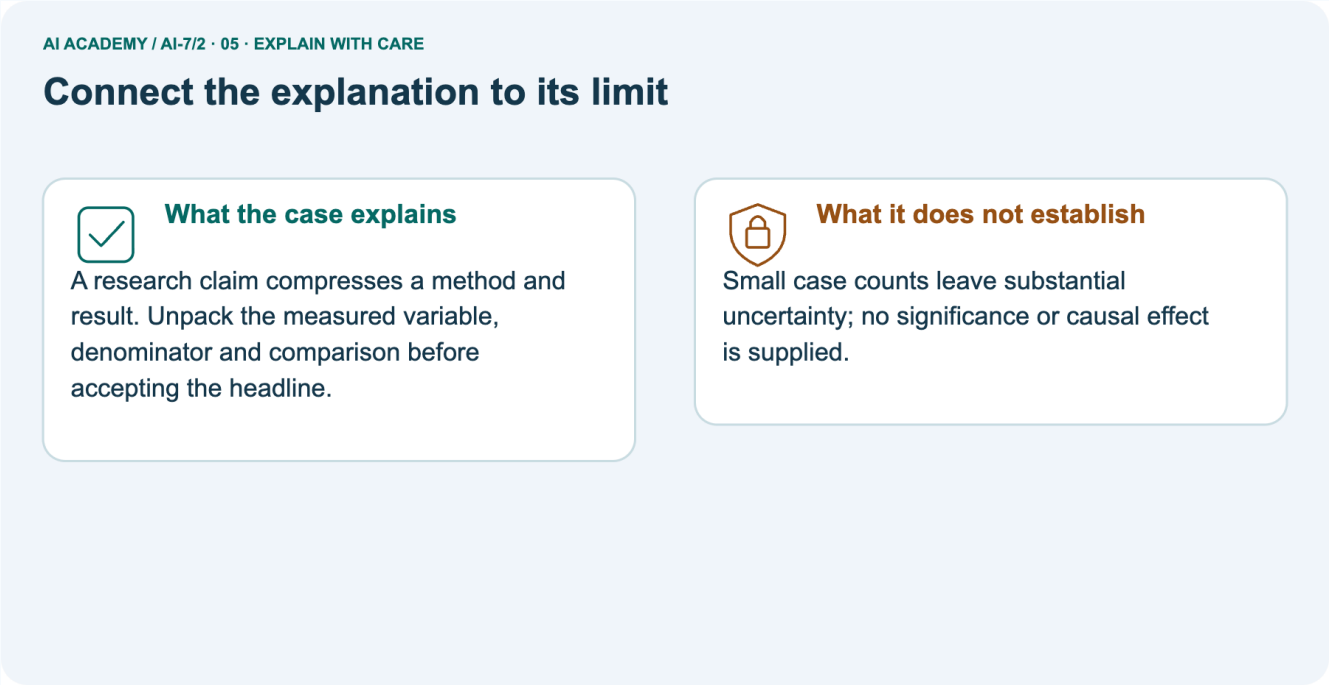
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: These ten cases support a descriptive comparison, not universal doubling. Changed condition: The headline changes to "every learner improves twice as much".

Expected result: That learner **claim** is unsupported by these answer counts. Reason: The outcome and experimental unit have changed in the wording without new **evidence**.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



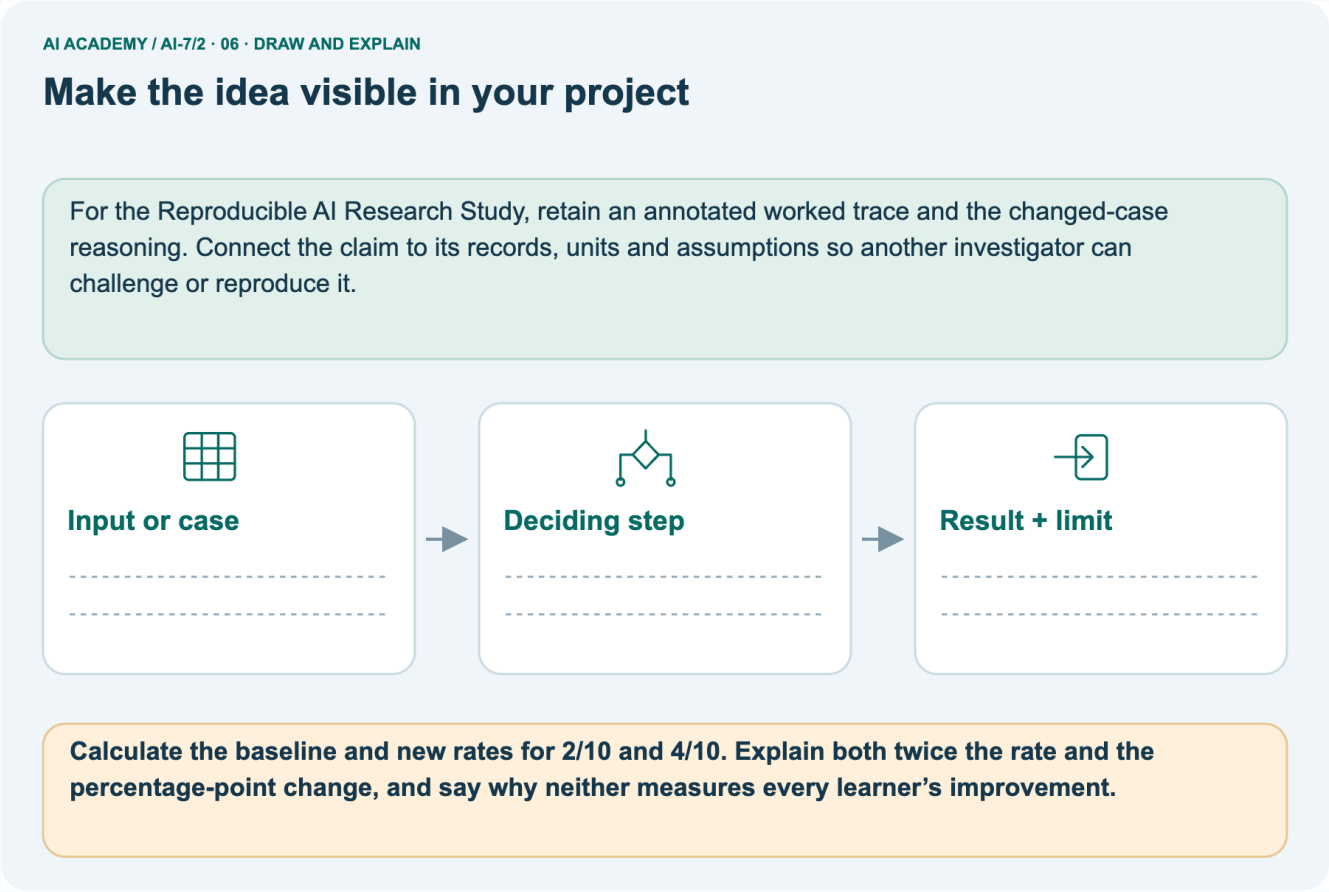
Follow the labelled picture with the matching explanation. Open the vector diagram

A research **claim** compresses a method and result. Unpack the measured variable, denominator and comparison before accepting the headline.

Limit: Small case counts leave substantial uncertainty; no significance or causal effect is supplied.

> Explain the case, then point to a limit on the **claim**.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Reproducible AI Research Study, retain an annotated worked trace and the changed-case reasoning. Connect the **claim** to its records, units and assumptions so another investigator can challenge or reproduce it.

Draw your **input**, deciding step, result and limit. Supported question: Calculate the baseline and new rates for 2/10 and 4/10.

Explain both twice the rate and the percentage-point change, and say why neither measures every learner's improvement.

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.